Paper 1

Exploring Embedding Interpretability by Correspondences Between Topic Models and Text Embeddings, 2025 (LINK)

- While dense vectors are inherently uninterpretable, topic models are (e.g. LDA and NMF). They analyze a corpus of documents, and output a list of human readable topics
    - For example, the words [algorithm, memory, cache] could define a topic
- Methodology:
    - The authors generated dense embeddings for a whole corpus of documents, as well as a topic model
    - They trained a linear transformation (projection matrix) to map the embedding space onto the topic distribution space
    - Thus the axes of the vector are human-readable topics, and so a high value on a particular axis means strong alignment with a certain topic
- Measurement of interpretability:
    - They used the “Intruder Word Test”
    - They took a specific axis and the top words that activated it, then injected it with an intruder word that does not belong
    - If a human participant can easily spot the intruder word, it means the embedding axis cleanly corresponds to a semantic concept

Paper 2

Text Embeddings Reveal (Almost) As Much As Text, 2023 (LINK)

- This paper focuses on Embedding Inversion, or trying to recover the original text given an embedding vector.
- Previously, it was thought that this was not possible as it was thought that embeddings stored semantic meanings instead of lexical data (exact phrasing)
- Concept of “conditioning”:
    - When you send text into an embedding model, the input words “condition” the output
    - The researchers showed their decoder by feeding it millions of (vector, original text) pairs, and the model learned how to map the vector geometry back into the English vocabulary
    - The researchers then fed the decoder new unseen vectors, and the decoders were able to generate the original words back with high accuracy
- Metrics:
    - Exact string match rate
    - BLEU (Bilingual Evaluate Understudy), which measures the overlap of word sequences (n-grams) between the original and reconstructed text

Paper 3

Amnesic Probing: Behavioral Explanation with Amnesic Counterfactuals, 2021 (LINK)

- This paper introduces the concept of Amnesic Probing, where instead of focusing on whether or not specific data is present in the embedding vector, it measures if the downstream model/system cares if the data is there or not
- Iterative Null-Space Projection (INLP):
    - Train a linear classifier to detect a feature in the dense embedding vectors, where W is the weight matrix
    - Project into the null space of W, with the goal of stripping away the vector space direction that the classifier used to find the feature
    - Since neural networks are redundant, retrain a new classifier on the altered embeddings, and keep repeating the process until no classifier can detect the feature with accuracy better than guessing
    - Then once done, feed these embeddings into a downstream task and measure the performance drop delta (can use BLEU score for example)
- The authors cared about whether language model embeddings cared about (used) specific concepts like Parts of Speech, verb tense, word gender
- And they found that while mathematically present in the embeddings, downstream models performed similarly on tasks like next word prediction and subject-word agreement

Takeaways

- Paper 1 gives a good way to map dense embeddings to topic models we can interpret and provides a test to verify them
- Paper 2 provides a way to quantify data lost during embedding inversion
- Paper 3 provides a way to quantify the impact of data loss on a downstream system